

BINDING — Environment ≠ advancement; score ≠ select (Charles lock, Jul 2026)

Status: Binding for agents, re-entry docs, basketball v1, and manuscript framing.

Last sharpened: 2026-07-28 — **scoring** and **selection** (winner rule) are separate pipeline steps.

Charles #1 confusion resolved: Do **not** merge “describe the peer environment” with “who wins the scarce slot.”

Filename note: File kept as `BINDING_Selection_is_its_own_step.md` for stable links. “Selection is its own step” still means: advancement is **not** environment. Inside advancement, **score ≠ select**.

Three separations (keep all of them)

Separation	Keep apart	Objects
1. Environment vs advancement	Peer help/hurt vs who gets the scarce slot	<code>L_net = B - D</code> vs score + winner rule
2. Scoring vs selection	How we rank vs how we pick winners	<code>S_i</code> (λ) vs top K (later: stochastic draw)
3. Hero vs equations	Empirical outcome plot vs generative rules	Layer A curve ≠ Alex score

Environment vs advancement

	Environment / development	Advancement (who gets the slot)
Question	How do peers affect net value, learning, visibility?	Who gets the draft pick / promotion / slot?
Symbols (theory)	<code>L_net = B(·) - D(·)</code>	Score <code>S_i</code> , then a winner rule
What it is NOT	The full story of who gets drafted	The same as fitting the hero curve
Layer A hero	Shows outcome (draft rate vs pool quality)	Does not separate environment vs advancement channel

Scoring vs selection (inside advancement)

	Scoring	Selection (winner rule)
Question	How do we rank candidates?	Given ranks, who wins ?
Symbols / knobs	<code>S_i = A_i - λ·L_C</code> (Alex v1); λ lives here	Top K (v1 default); later soft / noisy / stochastic draw
Pass A knockout	Toggle congestion in the score ($\lambda=0$ vs $\lambda>0$)	Same winner rule held fixed
Future noise	Leave <code>S_i</code> fixed	Change only this step

One sentence: *The hero describes outcomes; `L_net` is the peer environment; advancement = **score** then **select**; the sim tests whether congestion **in the score** changes who gets selected under a fixed winner rule.*

Why Charles (and others) get lost

The project uses **B – D** for the **environment** and Alex’s `S_i` for **ranking**. If you treat environment + score + top-K as one blob (“the model”), you rebuild a complicated world where everything happens at once. If you call `S_i` “selection,” you cannot cleanly add stochastic draft later.

v1 simplification:

- **B – D** = prose frame for help vs hurt among peers (not separately estimated in v1).
- **Alex score = scoring only** — congestion enters the **ranking**, not “the whole economy.”
- **Winner rule = top K** for now (explicit, separate step).
- Generative code = make **score** → **select** visible: talent-only score vs congestion-in-score, same top-K.

What each layer owns

Layer	Owns
A	Empirical curve — <i>what</i> the data look like
B	Two-part story: (1) B vs D in environment; (2) D (congestion) may enter the score via Alex <code>S_i</code> , then a winner rule
C	Code: assign → score → select (top K) → plot

Agent rule

When explaining basketball model work to Charles or in manuscript prose:

1. Say **environment** when you mean B – D / peer effects on development.
2. Say **scoring** / **Alex score** / **S_i** when you mean the ranking equation (λ lives here).
3. Say **selection** / **winner rule** / **draft** when you mean top-K or a later stochastic pick from scores.
4. Never imply the hero regression **is** the scoring equation.
5. Layer C knockouts (Pass A) change **scoring ingredients**, not the winner rule and not every environmental channel.

Re-entry pointer: [re_entry/02_Three_Kinds_of_Model.md](#) — BINDING + scoring vs selection.